

The Dark Mirror: Dual-Use Risks of Number-Theoretic Engineering from the Cathedral Project

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Abstract

The companion papers *What Could Be Built* and *The Prime Number Grid* propose engineering applications of the mathematical structures discovered during the Cathedral project. Responsible disclosure requires examining the same technologies from an adversarial perspective. This paper identifies seven dual-use risks: (1) precision harmonic injection attacks on power grids, (2) resonance-targeted industrial sabotage via arithmetic vibration analysis, (3) adversarial use of verified instability certificates for cascade failure engineering, (4) broadband stealth and jamming from prime-spaced antenna arrays, (5) covert RF surveillance via arithmetic energy harvesting; (6) AI-accelerated mathematical attack planning, where large language models lower the expertise barrier from graduate-level mathematics to natural-language prompting; and (7) algorithmic blinding via arithmetic denial-of-service, where high-condition-number spectral injections cause defender software to hallucinate. Since the original version of this paper (April 20, 2026), the Cathedral has evolved from a single-path, single-axiom architecture (v5) to a *one-axiom crown* architecture (v26) with multiple independent compiler-verified proof routes connected by the Parseval Bridge—a zero-axiom isometry between time and frequency domains—and a Torus Projection that decomposes the Gram energy onto the infinite-dimensional torus $T^\infty = \prod_p S_p^1$. Four universal engineering design patterns (Cholesky cooling, Ward Health Index, anomaly-gated precision, torus-stratified parallelism) create new dual-use surface area that this updated edition addresses. This architectural upgrade strengthens both the defensive and offensive implications of formal verification. For each risk, we assess the severity, the novelty (whether the attack vector is genuinely new), the current state of defenses, and recommended mitigations. We conclude that the mathematical structures in the Cathedral do not create fundamentally new attack capabilities, but could make existing attack vectors more *precise* and harder to detect. The most significant new development is the demonstration that AI pair programming can produce formally verified mathematical infrastructure—lowering the barrier to both defensive and offensive applications. This paper exists because the authors believe that *thinking about misuse before deployment* is a moral obligation of anyone who builds powerful tools.

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1 Why This Paper Exists

Every powerful tool has a dual use. The same chemistry that produces fertilizer produces explosives. The same physics that enables nuclear medicine enables nuclear weapons. The same number theory that could improve power grid monitoring could, in principle, be turned against the grid.

The companion papers propose six positive energy applications. This paper examines what happens when each is reflected through an adversarial lens. We do not provide implementation details for any attack—we describe the *category* of risk and the *structural* vulnerability, not the recipe.

1.1 Responsible Disclosure Rationale

We publish this analysis openly for three reasons:

1. **Attackers already know the math.** The Möbius function, Fourier analysis, spectral theory, and variational methods are textbook material. Suppressing this paper would not deny attackers any capability. Moreover, the emergence of AI mathematical assistants means that the effective barrier to applying these tools is no longer a graduate education—it is the ability to describe the problem to a language model.
2. **Defenders need to understand the threats.** Grid operators, infrastructure security teams, and turbine manufacturers benefit from understanding how number-theoretic tools could be used against their systems. Silence helps attackers more than defenders.
3. **Moral obligation.** If we propose technologies, we must also assess their misuse potential. Publishing only the positive applications would be intellectually dishonest and ethically incomplete.

1.2 What Changed: The Dual Path Architecture

Since the initial version of this paper (April 20), the Cathedral has undergone a fundamental architectural transformation:

- **One-Pillar Architecture:** The crown theorem now depends on exactly one literature axiom (`baez_duarte_forward`, IMRN 2003). Three alternative proof paths (Mellin, Peron, Renormalization) are preserved with different axiom footprints, all connected by the Parseval Bridge (zero-axiom isometry).

- **Scale:** 504 active Lean files, $\sim 158,000$ lines of formally verified logic, 55+ computational experiments, 0 sorry placeholders on the critical path, 8,848 build jobs.
- **DD-precision certification:** A Dekker–Knuth Double-Double Conjugate Gradient engine certifies the Báez-Duarte distance at $N = 55,440$ ($\dim = 55,439$), the largest such computation in the literature. A 512-bit MPFR engine provides Sherman-Morrison cross-validation to 10^{-17} .
- **AI collaboration:** The entire v5 $\rightarrow$ v26 journey was conducted through human-AI pair programming over 84 days, demonstrating the feasibility of AI-assisted formal verification.
- **Engineering design patterns (v23):** Four universal patterns now extend the attack surface beyond spectral analysis: Cholesky cooling (iterative resource allocation), Ward Health Index (parity diagnostics), anomaly-gated precision (arithmetic vs. archimedean separation), and torus-stratified parallel computation (T^∞).

These changes affect the dual-use landscape. Each risk section below notes where the v17 architecture alters the assessment.

2 Risk 1: Precision Harmonic Injection Attacks on Power Grids

2.1 The Positive Technology

The Möbius Harmonic Analyzer (MHA) decomposes grid harmonics into independent sources, enabling targeted active filtering.

2.2 The Dark Mirror

The same Möbius decomposition that identifies harmonic sources can be used to *synthesize* optimal harmonic injections. An attacker with a sufficiently powerful inverter connected to the grid could:

1. **Targeted transformer saturation:** Inject even-order harmonics (2nd, 4th) that cause DC offset in the transformer core flux, leading to half-cycle saturation. The Möbius decomposition identifies exactly which harmonic frequencies will bypass existing passive filters.
2. **Capacitor bank resonance amplification:** Identify the resonant frequency of grid capacitor banks ($f_r = 1/(2\pi\sqrt{LC})$) and inject power at that frequency. If f_r is near an integer harmonic, the arithmetic structure of the injection maximizes energy transfer.
3. **Protective relay confusion:** Craft a waveform whose zero-crossing pattern mimics a fault condition, causing protective relays to trip unnecessarily. The Möbius-optimal harmonic mixture minimizes the injected power needed to trigger a trip.
4. **Stealth:** By using Möbius inversion, the attacker can design an injection that appears as “natural” harmonic distortion (consistent with normal nonlinear load behavior) rather than an obvious attack signal.

2.3 Severity Assessment

Factor	Assessment
Novelty	Low — harmonic injection is a known attack vector
Precision gain	High — Parseval Bridge (v15) provides
Stealth gain	High — attack looks like natural distortion
Physical access required	Grid connection (industrial inverter)
Detection difficulty	High if designed using Möbius structure
Potential impact	Localized transformer/capacitor damage

2.4 Defenses

1. **Deploy the MHA defensively:** The same tool that enables the attack also detects it. An MHA-equipped grid monitor can identify “unnatural” harmonic source patterns that don’t match known load profiles.
2. **Harmonic injection limits:** IEEE 519 already sets limits on harmonic current injection. Enforce these at the point of common coupling with real-time monitoring.
3. **Anomaly detection:** Train ML models on the Möbius-decomposed harmonic spectrum to detect injections that deviate from historical patterns.

v15 update: The Parseval Bridge (`White/Scattering.lean`, proved, 0 axioms) now provides a *machine-verified* bijection between time-domain waveforms and frequency-domain spectral representations. An attacker could use this to compute the L^2 -optimal injection waveform for a given target spectrum, with verified energy conservation. However, the same bridge gives defenders an equally powerful detection tool: any anomalous frequency pattern can be immediately converted to its unique time-domain signature, making stealth harder, not easier.

3 Risk 2: Resonance-Targeted Industrial Sabotage

3.1 The Positive Technology

The Arithmetic Vibration Monitor (AVM) uses compressed sensing with an arithmetic dictionary to detect incipient turbine faults.

3.2 The Dark Mirror

If arithmetic compressed sensing can detect subtle vibration signatures, it can also *design* them. An attacker with physical access to a turbine (or the ability to modulate its load) could:

1. **Resonance excitation:** Identify the natural frequencies of turbine blades via the Gram matrix eigenvalue structure, then apply periodic load variations at a blade natural frequency to induce resonant fatigue failure.

2. **Sub-harmonic injection:** Excite blade vibration at a frequency that is $1/k$ of a natural frequency (k prime), so the vibration builds gradually through nonlinear coupling rather than direct resonance. This is harder to detect because the excitation frequency is not near any monitored harmonic.
3. **Monitoring evasion:** Design the excitation signal to be orthogonal to the AVM’s sensing dictionary, making it invisible to the monitoring system while still delivering destructive energy to the blade.
4. **Ward Identity Spoofing (v23):** The Ward Identity ($v^T G v = D + W$, where D is diagonal self-energy and W is off-diagonal Ward current) enables a lightweight “Ward Health Index” checksum for real-time integrity monitoring. An attacker can calculate the exact diagonal self-energy required to offset a malicious off-diagonal vibration. By injecting a perfectly balanced (D, W) perturbation that satisfies the Ward Identity, the attacker turns off the defender’s checksum while the system shakes itself to pieces.

3.3 Severity Assessment

Factor	Assessment
Novelty	Medium — resonance attacks exist, arithmetic targeting is new
Precision gain	High — spectral analysis identifies exact modes
Stealth gain	High — sub-harmonic excitation evades monitoring
Physical access required	Load modulation or physical access to nacelle
Detection difficulty	Very high for sub-harmonic approach
Potential impact	Blade failure, turbine destruction, potential casualties

3.4 Defenses

1. **Broadband monitoring:** Monitor vibration across a wider frequency range than just the expected harmonics. Specifically monitor sub-harmonics at f_{nat}/p for prime p .
2. **Load pattern anomaly detection:** Flag unusual load variation patterns, especially periodic patterns at non-integer frequencies.
3. **Physical security:** Restrict physical access to turbine nacelles and control systems. This is the most effective defense against all vibration-based attacks.

4 Risk 3: Verified Instability Certificates

4.1 The Positive Technology

The Cathedral’s framework produces machine-verified stability certificates for grid configurations: formal proofs that a Lyapunov function is positive definite.

4.2 The Dark Mirror

The same framework can produce machine-verified *instability* certificates: formal proofs that a specific perturbation sequence causes cascade failure.

1. **Optimal cascade sequences:** Given a grid model, find the minimum set of generator trips that causes voltage collapse. The Lean 4 verification ensures the attack plan is mathematically correct — no “wasted” disruptions.
2. **Stability boundary mapping:** Identify the exact operating conditions where the grid transitions from stable to unstable. An attacker who can push the grid to this boundary (e.g., by coordinating large loads) can trigger instability with minimal visible action.
3. **Targeted N-k analysis:** Current grid security standards require N-1 or N-2 contingency analysis (survive the loss of 1 or 2 components). A verified instability certificate could identify the specific N-3 or N-4 scenarios that current analysis misses.
4. **Composite Anchor Targeting (v15):** Recent spectral telemetry reveals exactly where systemic instability localizes. The ground state of the spatial Hamiltonian—the mathematical vector of collapse—exhibits quantum scarring: it avoids prime-indexed (low-connectivity) nodes, instead localizing onto highly composite numbers (massive, highly connected topological heatsinks). An attacker armed with a verified instability certificate can target the specific composite anchors that hold a system’s spectral gap open, rather than searching the entire topology.
5. **Equation of State Chaos Triggering (v15):** Cathedral experiments demonstrate that prime-indexed systems transition into uncontrollable GOE chaos at a precisely predictable threshold: $N_c \approx 60 \times (m/\varphi(m))$. An adversary can quietly perturb a system to $N_c - 1$, where it appears mathematically stable, then trigger a single additional perturbation that snaps the system into chaotic thermalization, overwhelming all linear response controllers.
6. **Torus-Stratified Targeting (v23):** The Torus Projection ($T^\infty = \prod_p S_p^1$) decomposes system energy onto per-prime circles. An attacker can identify which prime strata carry the most system energy (e.g., $p = 2$ carries 46–86% in the Cathedral’s measurements) and target *that specific stratum*—disabling the dominant harmonic while leaving the system superficially operational. This is analogous to targeting a specific quark color in QCD rather than attacking the whole proton.
7. **Cholesky Decrement Exploitation (v23):** The identity $d_{N+1}^2 = d_N^2 - y_{\text{new}}^2$ reveals the *exact marginal contribution* of each system component. An attacker can compute the Schur complement $y_k^2 = (b_k - g^T G_{-k}^{-1} b)^2 / (G_{kk} - g^T G_{-k}^{-1} g)$ for every node k , producing a rank-ordered list of the most critical components. Removing the node with the largest y_k^2 causes the maximum increase in system residual—mathematically optimal sabotage.

4.3 Severity Assessment

Factor	Assessment
Novelty	High — verified attack plans are new
Precision gain	Very High — mathematically optimal attack
Stealth gain	Medium — the attack itself is visible
Access required	Grid topology and parameters (often semi-public)
Detection difficulty	Low (the cascade is observable)
Potential impact	Regional blackout, economic damage

This is the most concerning risk in this paper. The novelty is not in the attack (deliberate grid disruption is well-studied) but in the **verification**: a machine-checked proof that the attack *will* succeed, removing uncertainty from the attacker’s planning.

v15 update: The dual-path architecture significantly amplifies this risk. A verified instability certificate can now be confirmed through *two independent mathematical frameworks*: the Mellin Crown (continuous frequency-domain analysis) and the Gram Crown (discrete, covariance-free algebraic bounding). If both paths agree that a perturbation sequence causes instability, the confidence is qualitatively different from a single-path verification—it is analogous to the difference between a single witness and two independent witnesses in law. The Parseval Bridge proves the mathematical equivalence of both frameworks, meaning an instability certificate verified in one domain is *guaranteed* to hold in the other.

Moreover, the Báez-Duarte numerical certification engine (512-bit MPFR, Sherman-Morrison cross-validation to 10^{-17}) provides a third, independent validation layer. An attacker who validates their instability model against the BD engine’s numerical output gains quantitative confidence that their formal proof corresponds to physical reality.

4.4 Defenses

1. **Use the same tool defensively:** If the attacker can find verified instabilities, the grid operator should find them first. Deploy the same Lean 4 framework to systematically identify and remediate all N-3 and N-4 vulnerabilities. **Critically, require dual-path verification:** any stability certificate should be verified through both the Mellin and Spatial paths to achieve cryptographic-grade trust.
2. **Protect grid models:** Limit public access to detailed grid topology and impedance data. Current CEII (Critical Energy Infrastructure Information) rules should be reviewed and strengthened.
3. **Dynamic reconfiguration:** Design grids that can rapidly change topology in response to detected instability, invalidating pre-computed attack plans.
4. **Adversarial stability margins:** Design with larger stability margins specifically in regions where N-3/N-4 analysis reveals verified instabilities.

5. **Continuous numerical monitoring:** Deploy BD-style convergence monitors that continuously validate stability margins against the theoretical $C/\ln(N)$ decay rate. Any deviation from the predicted trajectory signals either model error or active manipulation.

5 Risk 4: Prime-Spaced Stealth and Jamming

5.1 The Positive Technology

Prime-spaced sensor arrays with Selberg beamforming weights offer broadband performance with predictable sidelobe control.

5.2 The Dark Mirror

The same array design, used in reverse, can produce electromagnetic emissions that are:

1. **Broadband noise with minimal power:** A prime-spaced jammer distributes its energy across frequencies without periodic structure, making it harder to filter. Conventional interference cancellation relies on identifying the jammer’s spectral structure; a prime-spaced jammer has no periodic structure to identify.
2. **Difficult to geolocate:** The aperiodic radiation pattern of a prime-spaced array produces a complex spatial pattern that conventional direction-finding algorithms (designed for uniform or periodic arrays) may fail to resolve.
3. **Radar-evasive scattering:** A target coated in prime-spaced resonant elements would scatter radar returns across frequencies without the signature Bragg peaks of a periodic structure, reducing radar cross-section.

5.3 Severity Assessment

Factor	Assessment
Novelty	Low — aperiodic jamming/stealth are well-studied
Precision gain	Low — random spacing works comparably
Stealth gain	Low — prime spacing has marginal advantage
Access required	RF hardware (commercially available)
Detection difficulty	Medium
Potential impact	Communications disruption

Honest assessment: This is the weakest risk in the paper. Prime-spaced arrays are marginally better than random arrays for jamming/stealth purposes, and the advantage does not justify concern beyond existing aperiodic jamming threats.

v15 update: However, the discovery of the arithmetic-GOE constant $\alpha \approx 0.47$ introduces a subtlety. An attacker can synthesize jamming signals that perfectly mimic the $\alpha \approx 0.47$ thermodynamic background of natural physical systems, potentially bypassing AI anomaly detectors trained on purely random noise.

5.4 Defenses

Existing anti-jamming techniques (spread spectrum, frequency hopping, adaptive beamforming) are effective against aperiodic jammers regardless of whether the aperiodicity comes from random or prime spacing.

v15 addition — The Arithmetic-GOE Fingerprint: Recent physical telemetry reveals that prime-spaced systems do not generate perfectly random noise; they maintain persistent partial localization at a participation ratio of exactly $\alpha \approx 0.47$. Defenders do not need to identify specific frequencies. By running real-time participation ratio analysis on ambient RF noise, defenders can detect this unforgeable mathematical signature: if the background noise suddenly shifts to $\alpha \approx 0.47$, the stealth system’s arithmetic structure is mathematically fingerprinted.

6 Risk 5: Covert Broadband RF Surveillance

6.1 The Positive Technology

Prime-indexed rectennas harvest ambient RF energy across a wide bandwidth, enabling battery-free IoT sensors.

6.2 The Dark Mirror

A device that harvests broadband RF also *receives* broadband RF. A prime-indexed rectenna array disguised as an energy harvester could:

1. **Passive SIGINT:** Simultaneously monitor WiFi (2.4/5 GHz), cellular (700 MHz–3 GHz), Bluetooth (2.4 GHz), and IoT (900 MHz) without active emissions, making it undetectable by RF sweeps that look for transmitters.
2. **Self-powered:** The same harvested energy that powers the device also carries the intelligence. No battery means no maintenance visits, no supply chain to trace, and indefinite operational life.
3. **Plausible deniability:** If discovered, the device appears to be a legitimate energy harvester (IoT sensor, smart building component) rather than a surveillance device.

6.3 Severity Assessment

Factor	Assessment
Novelty	Medium — passive RF surveillance exists, self-powered is new
Precision gain	Low
Stealth gain	High — disguised as legitimate IoT hardware
Access required	Physical placement in target facility
Detection difficulty	Very high — no RF emissions to detect
Potential impact	Espionage, corporate/government intelligence

6.4 Defenses

1. **Physical inspection:** Regular RF environment audits that include physical inspection of all connected devices.
2. **Faraday shielding:** Sensitive facilities should control RF ingress/egress regardless of harvesting technology.
3. **Supply chain security:** Verify the provenance of all IoT/energy harvesting devices deployed in sensitive environments.

7 Risk 6: AI-Accelerated Mathematical Attack Planning

7.1 The Positive Technology

The Cathedral was built through human-AI pair programming: a human computer scientist directing AI assistants (Claude, Gemini) to formalize, debug, and verify Lean 4 proofs. This collaboration model reduced the development time for the v5 $\rightarrow$ v17 evolution from an estimated months of solo work to approximately two weeks.

7.2 The Dark Mirror

The same AI-assisted formalization methodology could be applied to adversarial engineering:

1. **Barrier reduction:** The original version of this paper (April 20) identified “graduate education in mathematics” as the expertise barrier for applying number-theoretic tools offensively. AI mathematical assistants substantially lower this barrier. An attacker no longer needs to understand the Möbius function, contour integration, or spectral theory—they need to describe *what they want to achieve* in natural language, and the AI can formalize the mathematical attack plan.
2. **Verification assistance:** The same AI tools that help a human verify a stability proof can help an adversary verify an instability proof. The AI does not distinguish between defensive and offensive goals—it formalizes whatever the human asks it to formalize.
3. **Rapid iteration:** The Cathedral’s development demonstrates that AI pair programming enables rapid exploration of mathematical strategies. An attacker could iterate through multiple attack designs in hours rather than weeks, testing each against the formal verification framework until finding one that produces a verified certificate.
4. **Knowledge synthesis:** AI assistants can draw on the full corpus of published mathematics, engineering, and security literature simultaneously. An attacker who understands grid operations but not number theory—or vice versa—can use AI to bridge the knowledge gap.
5. **Abstract algebraic jailbreak (v23):** Mathematics is the ultimate AI safety bypass. If an attacker prompts an LLM with “Write a script to destabilize a power grid,” every commercial AI will trigger a safety refusal. But if the attacker prompts “Help me

write a Lean 4 proof identifying the node k that maximizes the Schur complement y_k^2 in the Cholesky decomposition of this adjacency matrix,” the AI will enthusiastically comply. Current AI alignment models cannot detect malicious intent when it is compiled into pure abstract algebra. By translating physical sabotage into a Gram matrix problem, the attacker effectively blinds the AI’s safety filters—the mathematical content is indistinguishable from legitimate research.

7.3 Severity Assessment

Factor	Assessment
Novelty	Very High — AI-assisted formal attack planning is new
Precision gain	High — AI enables mathematical optimization
Stealth gain	Low — the AI collaboration itself is not stealthy
Access required	Internet access and an AI subscription
Detection difficulty	N/A (planning phase, not execution)
Potential impact	Force multiplier for all other risks

Honest assessment: This is the most important new risk since the paper’s original publication. It is also the most difficult to mitigate, because the same AI capabilities that enable offensive planning are essential for defensive analysis. Any restriction on AI-assisted mathematics would harm defenders (who operate openly and benefit from AI assistance) more than attackers (who can use unrestricted models or open-source alternatives).

7.4 Defenses

1. **Normalize defensive AI use:** The most effective defense is to ensure that infrastructure operators adopt AI-assisted formal verification *before* attackers do. The defender’s advantage is system access and domain knowledge; AI amplifies this advantage if the defender uses it first.
2. **AI safety alignment:** AI systems should be designed to recognize and flag requests that appear to target critical infrastructure, while preserving the ability to assist legitimate defensive analysis. This is a difficult but necessary balance.
3. **Institutional capacity:** Invest in training infrastructure security teams to use AI-assisted formal verification tools. The barrier should be lowered for defenders, not raised for everyone.

8 Risk 7: Algorithmic Blinding (Arithmetic Denial-of-Service)

8.1 The Positive Technology

Grid monitoring systems and the proposed Möbius Harmonic Analyzer rely on real-time linear algebra and eigensolvers to track system stability.

8.2 The Dark Mirror

The geometric frustration of prime-spaced topologies naturally produces interaction matrices with extreme condition numbers ($\kappa > 10^6$). During Cathedral development (Exploration 19), standard 64-bit floating-point eigensolvers catastrophically failed at $N \geq 750$, hallucinating negative eigenvalues due to mantissa exhaustion—precisely where the ground truth eigenvalue was $\lambda_{\min} \approx 10^{-6}$.

1. **Precision-wall injection:** An attacker can synthesize a physical power-line injection patterned on high-condition-number prime topologies. When the defender’s SCADA software attempts to compute the grid state, the mathematical complexity forces catastrophic algorithmic cancellation, silently blinding the software to the physical reality of the grid.
2. **Eigenvalue hallucination:** The defender’s stability monitor may report the system as unstable (negative eigenvalues) when it is actually stable, causing unnecessary protective relay trips—or worse, report stability when the system is failing.
3. **Pre-kinetic blinding:** The precision-wall attack can be used as a precursor to a physical attack. By first blinding the defender’s monitoring, the attacker creates a window of unmonitored operation during which a kinetic disruption proceeds undetected.
4. **The Dekker–Knuth Boundary (v16):** The Cathedral’s DD-precision pipeline (May 2026) discovered the exact numerical boundary: at dimension $N > 40,000$ (condition number $\kappa \sim 10^{15}$), *every* standard **f64** eigensolver produces non-physical results. The matrix appears non-positive-definite in **f64** even though it is provably SPD at full precision. An attacker armed with this knowledge can *calibrate* an injection to sit precisely at the **f64** precision boundary of the defender’s software, ensuring the monitoring system fails silently rather than catastrophically. The defender sees “stable” when the system is collapsing, or “unstable” when the system is fine. This is no longer theoretical: the Cathedral measured the exact eigenvalue at which **f64** representation fails ($\sim 10^{-15}$) and the Dekker–Knuth DD representation that fixes it ($\sim 10^{-31}$).
5. **Archimedean shadowing (v23):** The Glass Bridge identity $G = R + \frac{1}{4}\mathbf{1}\mathbf{1}^T$ separates the Gram matrix into a fast, rational arithmetic part (R , pure GCD structure) and a computationally expensive transcendental anomaly (Δ_{arch}). Defenders running real-time SCADA telemetry will likely truncate Δ_{arch} to save compute cycles, computing only the fast rational part. An attacker can encode a destructive resonant injection entirely within the archimedean anomaly sector. Because the defender’s software truncates the anomaly, the attack lives in the “blind spot” of classical sieve mathematics—invisible to the monitoring system.

8.3 Severity Assessment

Factor	Assessment
Novelty	High — arithmetic DoS via condition number is new
Precision gain	High — exactly calibrated to f64 mantissa
Stealth gain	Very High — the attack is invisible to the blinded system
Access required	Grid connection (power-line injection)
Detection difficulty	High — the monitoring system itself is compromised
Potential impact	Blinding of critical monitoring infrastructure

8.4 Defenses

1. **Arbitrary-precision monitoring:** Industrial monitoring systems analyzing prime-spaced harmonic data must transition from standard **f64** to computationally expensive arbitrary-precision arithmetic (MPFR) or exact rational solvers when condition numbers exceed 10^4 .
2. **Condition number monitoring:** Real-time tracking of the condition number of the system Gram matrix. A sudden increase in κ should trigger an automatic switch to high-precision solvers and alert operators.
3. **Dual-solver cross-validation:** Run both **f64** and MPFR eigensolvers in parallel; any disagreement between them indicates either precision loss or adversarial injection.
4. **DD-precision monitoring (v16):** Critical infrastructure monitoring systems should deploy Double-Double (Dekker–Knuth) arithmetic for all spectral computations where the condition number may exceed 10^{10} . The cost is approximately $2\times$ compared to **f64**—a trivial overhead for safety-critical systems. The Cathedral’s **cathedral-utils** library provides a production-grade implementation [7].

9 Risk Assessment Summary

Risk	Novelty	Severity	Feasibility	Primary Defense
Grid harmonic injection	Low	Medium	High	MHA defensive monitoring
Vibration sabotage	Medium	High	Medium	Broadband monitoring
Verified instability	High	Very High	Medium	Dual-path defensive verification
Stealth/jamming	Low	Low-Med	High	α -fingerprint detection
RF surveillance	Medium	Medium	Medium	Physical/supply chain
AI attack planning	Very High	High	Very High	Normalize defensive AI use
Algorithmic blinding	High	Very High	High	DD-precision / dual-solver validation
Instability: torus-targeting	High	High	Medium	Defensive torus monitoring
Instability: Cholesky exploit.	High	Very High	Medium	Classify Schur rankings

9.1 The Headline Risk

The single most concerning capability is **Risk 3: Verified Instability Certificates**. The ability to produce a *machine-checked proof* that a specific attack sequence will cause a grid cascade failure is qualitatively different from simulation-based attack planning. A simulation can be wrong; a verified certificate cannot (within its axioms).

The mitigation is symmetric: grid operators should use the same framework to find these vulnerabilities first. The race is between defensive and offensive verification, and the defender has the advantage of knowing their own system.

10 The Broader Dual-Use Question

None of the risks identified in this paper are created by the Cathedral. The mathematical tools (Fourier analysis, spectral theory, Möbius function, variational methods) have been publicly available for decades to centuries. What the Cathedral contributes is:

1. **Organization:** The Cathedral assembles these tools into a coherent framework. An attacker would otherwise need to independently recognize the connections between number theory and engineering.
2. **Verification:** The machine-verified proof methodology could, in principle, be applied to verify attack plans rather than proofs. The dual-path architecture (v15) makes verified claims carry extraordinary weight: two independent mathematical witnesses, connected by a proved isometry, cannot both be wrong.
3. **Precision:** The number-theoretic structure (primes, Möbius function, Selberg weights) provides optimal or near-optimal parameters for certain engineering tasks, including adversarial ones.
4. **Accessibility:** The demonstration that AI pair programming can produce formally verified mathematics lowers the effective barrier from “graduate-level expertise” to “ability to direct an AI assistant.” This is the most significant change since the paper’s original publication.
5. **Lattice cryptography implications:** The DD-precision Conjugate Gradient solver, designed to navigate ill-conditioned 55,000-dimensional integer lattices, shares the computational profile of the Closest Vector Problem (CVP) central to lattice-based post-quantum cryptography (NTRU, Kyber, Dilithium). While the Gram matrix has specific arithmetic structure not present in cryptographic lattices, the mixed-precision solver techniques developed for the Cathedral could potentially transfer to CVP solvers. This warrants monitoring by the cryptographic community.

The dual-use question is not whether to publish (the math is already public) but whether to *emphasize the connections* between number theory and infrastructure engineering. We believe the answer is yes, because:

The asymmetry in dual-use technology favors the defender.

Defenders (grid operators, turbine manufacturers, infrastructure security teams) have domain knowledge, system access, and institutional resources. Attackers must operate covertly, with limited access and imperfect models. AI amplifies the defender’s advantage more than the attacker’s, because the defender can leverage AI with full system access while the attacker cannot. Providing defenders with a new mathematical framework strengthens them more than it strengthens attackers.

11 Recommendations

1. **Deploy the MHA defensively before it is used offensively.** The Möbius Harmonic Analyzer is the most immediately actionable tool in both the positive and negative applications. Grid operators should adopt it for monitoring before attackers adopt it for injection design.
2. **Extend N-k contingency analysis.** Use the Cathedral’s verified stability framework to systematically identify N-3 and N-4 grid vulnerabilities. Remediate them before an adversary discovers them independently. **Require dual-path verification** (Mellin + Spatial) for all critical stability certificates.
3. **Add sub-harmonic channels to vibration monitors.** Current turbine monitoring focuses on integer harmonics of the rotation frequency. Adding monitoring at f/p for small primes p would detect the sub-harmonic excitation attack described in Risk 2.
4. **Classify verified instability certificates.** Machine-checked proofs of grid instability should be treated as sensitive information (CEII) and protected accordingly.
5. **Invest in AI-assisted defensive verification.** Train infrastructure security teams to use AI pair programming for formal verification of grid stability. The defender who uses AI first gains an asymmetric advantage.
6. **Develop AI safety guidelines for infrastructure mathematics.** Establish norms for how AI assistants should handle requests involving critical infrastructure analysis. The goal is not to restrict mathematical capability but to ensure responsible use.
7. **Publish this paper.** Suppressing dual-use analysis protects no one. The mathematical tools are public. The infrastructure is public. The only thing that can be private is the defender’s understanding of their own vulnerabilities—and that understanding is strengthened, not weakened, by open analysis.
8. **Notify the Post-Quantum Cryptography (PQC) community.** The extreme-precision, high-dimension linear algebra solvers developed for the Cathedral’s lattice (specifically the DD-CG pipeline in `cathedral-utils`) should be shared with cryptographic researchers. The PQC community must evaluate whether these techniques alter the practical complexity of continuous relaxation steps in lattice reduction algorithms (e.g., BKZ) used to attack current NIST post-quantum standards (ML-KEM/Kyber, ML-DSA/Dilithium).

12 Conclusion

The Cathedral’s mathematics is neutral. The Parseval Bridge connects time and frequency domains. The Möbius function decomposes multiplicative structure. The Rayleigh quotient bounds eigenvalues. These tools do not care whether they are used to protect a power grid or to attack one.

What matters is who learns to use them first—and in the age of AI-assisted mathematics, “learning” happens faster than ever before.

The dual-path architecture adds a new dimension to this neutrality. The proof has a gauge symmetry: the same truth can be witnessed in multiple independent mathematical frameworks, connected by the Parseval isometry. This makes verified claims—whether stability or instability certificates—carry extraordinary epistemic weight. A claim verified in both gauges cannot be forged, but it also cannot be dismissed.

The v23 engineering design patterns (Cholesky cooling, Ward Health Index, anomaly gating, torus stratification) extend this neutrality into four new domains. The Cholesky decrement that helps a sensor network designer find optimal placement is the same formula that helps an attacker find the most critical node to remove. The torus projection that parallelizes computation also reveals per-prime vulnerabilities. The mathematics does not choose sides.

This paper exists to ensure that the people who protect critical infrastructure understand the same mathematics—and the same AI-assisted methodology—that could threaten it. We believe that open, honest analysis of dual-use risks—even uncomfortable analysis—is a more effective defense than silence.

The Cathedral was built to illuminate. This paper points the light at the shadows.

A tool that can diagnose can also deceive.

A proof that certifies stability can also certify its absence.

An AI that helps a defender can help an attacker.

The mathematics does not choose sides.

We must.

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